

Bo Ai

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RESEARCH INTERESTS

My research focuses on developing **collective robot intelligence**: robotic agents that transfer knowledge across diverse embodiments, collaborate with one another and with humans, and co-evolve their physical forms and capabilities over time. Concrete research questions include how to represent embodiment structure in robot policies, how transfer scales with embodiment diversity, and which hierarchical policy architectures enable effective multi-agent coordination.

EDUCATION

Stanford University

Beginning Sep 2026

Incoming Ph.D. student in Computer Science

Research interests: cross-embodiment transfer, multi-agent collaboration, and continual learning

University of California San Diego (UCSD)

Sep 2024 – Sep 2026

Ph.D. student in Computer Science and Engineering; M.S. awarded. Advisors: [Hao Su](#) and [Henrik I. Christensen](#)

Research interests: cross-embodiment transfer, world models, and robotic manipulation

National University of Singapore (NUS)

Aug 2019 - June 2023

B.Comp. in Computer Science with a Second Major in Statistics, Highest Distinction

Turing Program. Advisor: [David Hsu](#)

Thesis: Scaling Robot Learning: Generalization Through Invariant Representations

EXPERIENCE

Physical Intelligence (Pi)

Oct 2025 – April 2026

Research Intern with [Allen Ren](#) and [Quan Vuong](#)

San Francisco, CA

- Initiated and led research on cross-embodiment transfer at Pi.

- Developed pretraining and post-training recipes to improve transfer to novel robot embodiments.

Boston Dynamics AI Institute

June 2025 – Sep 2025

Research Intern with [Pang Tao](#) and [Jiuguang Wang](#)

Boston, MA

- Developed algorithms and systems for whole-body, contact-rich manipulation on the Spot robot.

Agency for Science, Technology and Research (A*STAR)

Dec 2023 – Sep 2024

Research Engineer with [Cheston Tan](#)

Singapore

- Led projects in world model learning and large language model reasoning.

Stanford Vision and Learning Lab, Stanford University

June 2023 – Dec 2023

Research Intern with [Yunzhu Li](#) and [Jiajun Wu](#)

Stanford, CA

- Developed a framework that learns world models with visual and tactile sensing from real-world interactions and integrates the learned models with planning for manipulating objects with unknown physical properties.

- Demonstrated the manipulation system to distinguished lab visitors.

Adaptive Computing Lab, National University of Singapore

Dec 2020 – May 2023

Undergraduate Research Assistant with [David Hsu](#)

Singapore

- Developed learning-based controllers and robotic systems that enable kilometer-scale visual navigation.

- Demonstrated systems to renowned faculty, senior industry executives, and government officials during lab visits.

SELECTED PUBLICATIONS AND PREPRINTS

[1] Physical Intelligence Team

$\pi_{0.7}$: A Steerable Generalist Robotic Foundation Model with Emergent Capabilities

Preprint, 2026

[2] Bo Ai, Stephen Tian, Haochen Shi, Yixuan Wang, Tobias Pfaff, Cheston Tan, Henrik I. Christensen, Hao Su, Jiajun Wu, Yunzhu Li

A Review of Learning-Based Dynamics Models for Robotic Manipulation

Science Robotics, 2025

[3] Bo Ai*, Liu Dai*, Nico Bohlinger*, Dichen Li*, Tongzhou Mu, Zhanxin Wu, K. Fay, Henrik I. Christensen, Jan Peters, Hao Su

Towards Embodiment Scaling Laws in Robot Locomotion

Conference on Robot Learning (CoRL), 2025

- [4] Zihao He*, **Bo Ai***, Tongzhou Mu, Yulin Liu, Weikang Wan, Jiawei Fu, Yilun Du, Henrik I. Christensen, and Hao Su
Scaling Cross-Embodiment World Models for Dexterous Manipulation
International Conference on Intelligent Robots and Systems (**IROS**), 2025
- [5] **Bo Ai***, Stephen Tian*, Haochen Shi, Yixuan Wang, Cheston Tan, Yunzhu Li, Jiajun Wu
RoboPack: Learning Tactile-Informed Dynamics Models for Dense Packing
Robotics: Science and Systems (**RSS**), 2024

SELECTED AWARDS AND HONORS

- Stanford School of Engineering Fellowship 2026
- UCSD CSE Department Fellowship 2024
- NUS School of Computing Dean's List (5%, Department-Wide) 2022
- NUS Outstanding Undergraduate Researcher Prize (35 Recipients Annually, University-Wide) 2022
- NUS School of Computing Innovation Prize (2 Recipients Annually, Department-Wide) 2022
- Certificate of Distinction in Artificial Intelligence 2022
- NUS Science and Technology Scholarship (~ 200K USD) 2018 - 2023

MENTORING

- Shresth Grover (MSc at UCSD → Intern at Princeton) 2025-2026
- Dichen Li (MSc at UCSD → PhD at Fudan University) 2024 - 2026
- Zihao He (Undergraduate at SJTU → PhD at UMD) 2025
- Chen Si (MSc at UCSD → PhD at NTU Singapore) 2025
- Tongxuan Tian (MSc at UCSD → PhD at USC) 2024 - 2025

TEACHING

- Guest Lecturer, UCSD CSE276F Machine Learning for Robotics *Spring 2025*
- Teaching Assistant, NUS CS5478 Intelligent Robots: Algorithms and Systems *Spring 2023*
- Teaching Assistant, NUS CS3244 Machine Learning *Fall 2022*
- Teaching Assistant, NUS CS1101S Programming Methodology *Fall 2020*

INVITED TALKS

- Integrating Learning and Planning for Robot Navigation**
– Stanford Intelligent Systems Lab *July 2023*
- Deep Learning for Robot Navigation with a Floor Map**
– NUS Smart System Institute *Feb 2022*

PROFESSIONAL ACTIVITIES

- Journal Reviewer:* IEEE RA-L '22-'25; IEEE RAM '25
- Conference Reviewer:* ICRA '22-'26; IROS '22-'24; CoRL '25-'26; RSS '26; CVPR '26; ECCV '26; NeurIPS '26

SKILLS

- Programming:** Python, C, Java, JavaScript, TypeScript, Elixir, HTML, LaTeX, R, SAS
- Robotics & Simulation:** ROS, ROS2, Isaac Lab, MuJoCo
- Robots:** Boston Dynamics Spot, Unitree Go2, Franka Panda, UR5e, ARX, XHand, LEGO Mindstorms, Arduino
- Languages:** English (fluent), Chinese (native)